

Администрирование локальных сетей

Статическая маршрутизация в Интернете. Настройка

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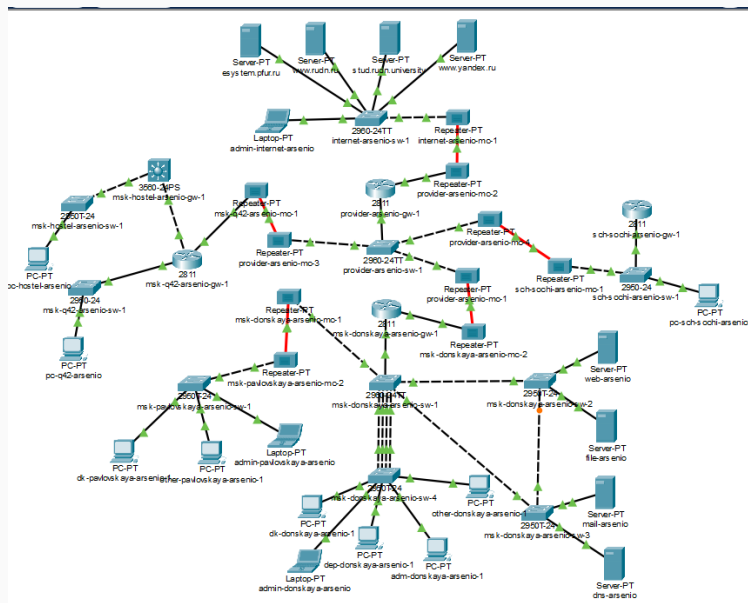
Российский университет дружбы народов, Москва, Россия

Цели и задачи работы

Настроить взаимодействие через сеть провайдера посредством статической маршрутизации локальной сети организации с сетью основного здания в 42-м квартале в Москве и сетью филиала в г. Сочи.

Топология сети

Общая схема сети



Настройка связи между территориями

VLAN на коммутаторе провайдера

```
provider-arsenio-sw-1>en
Password:
provider-arsenio-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
provider-arsenio-sw-1(config)#
provider-arsenio-sw-1(config)#vlan 5
provider-arsenio-sw-1(config-vlan)#name q42
provider-arsenio-sw-1(config-vlan)#int vlan 5
provider-arsenio-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan5, changed state to up

provider-arsenio-sw-1(config-if)#no sh
provider-arsenio-sw-1(config-if)#vlan 6
provider-arsenio-sw-1(config-vlan)#name sochi
provider-arsenio-sw-1(config-vlan)#int vlan 6
provider-arsenio-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up

provider-arsenio-sw-1(config-if)#no sh
provider-arsenio-sw-1(config-if)#^Z
provider-arsenio-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-arsenio-sw-1#wr mem
Building configuration...
[OK]
provider-arsenio-sw-1#
```

Маршрутизатор msk-donskaya-arsenio-gw-1

```
Password:
msk-donskaya-arsenio-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-arsenio-gw-1(config)#int f0/1.5
msk-donskaya-arsenio-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up

msk-donskaya-arsenio-gw-1(config-subif)#enc dot1q 5
msk-donskaya-arsenio-gw-1(config-subif)#ip addr 10.128.255.1 255.255.255.252
msk-donskaya-arsenio-gw-1(config-subif)#descr q42
msk-donskaya-arsenio-gw-1(config-subif)#int f0/1.6
msk-donskaya-arsenio-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.6, changed state to up

msk-donskaya-arsenio-gw-1(config-subif)#enc dot1q 6
msk-donskaya-arsenio-gw-1(config-subif)#ip addr 10.128.255.5 255.255.255.252
msk-donskaya-arsenio-gw-1(config-subif)#descr sochi
msk-donskaya-arsenio-gw-1(config-subif)#exit
msk-donskaya-arsenio-gw-1(config)#ip route 10.129.0.0 255.255.0.0 10.128.255.2
msk-donskaya-arsenio-gw-1(config)#ip route 10.130.0.0 255.255.0.0 10.128.255.6
msk-donskaya-arsenio-gw-1(config)#int f0/1.5
msk-donskaya-arsenio-gw-1(config-subif)#ip nat inside
msk-donskaya-arsenio-gw-1(config-subif)#int f0/1.6
msk-donskaya-arsenio-gw-1(config-subif)#ip nat inside
msk-donskaya-arsenio-gw-1(config-subif)#exit
msk-donskaya-arsenio-gw-1(config)#ip access-list ext nat-inet
msk-donskaya-arsenio-gw-1(config-ext-nacl)#remark q42
msk-donskaya-arsenio-gw-1(config-ext-nacl)#permit ip host 10.129.0.200 any
msk-donskaya-arsenio-gw-1(config-ext-nacl)#permit ip host 10.129.128.200 any
msk-donskaya-arsenio-gw-1(config-ext-nacl)#remark sochi
msk-donskaya-arsenio-gw-1(config-ext-nacl)#permit ip host 10.130.0.200 any
msk-donskaya-arsenio-gw-1(config-ext-nacl)#exit
msk-donskaya-arsenio-gw-1(config)#^Z
msk-donskaya-arsenio-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-arsenio-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-arsenio-gw-1#
```


Настройка квартала 42

Сабинтерфейсы маршрутизатора квартала 42

```
Password:
msk-q42-arsenic-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-arsenic-gw-1(config)#int f0/1
msk-q42-arsenic-gw-1(config-if)#no sh

msk-q42-arsenic-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

msk-q42-arsenic-gw-1(config-if)#int f0/1.5
msk-q42-arsenic-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up

msk-q42-arsenic-gw-1(config-subif)#enc dot1q 5
msk-q42-arsenic-gw-1(config-subif)#ip addr 10.128.255.2 255.255.255.252
msk-q42-arsenic-gw-1(config-subif)#descr donskaya
msk-q42-arsenic-gw-1(config-subif)#int f0/0
msk-q42-arsenic-gw-1(config-if)#no sh

msk-q42-arsenic-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

msk-q42-arsenic-gw-1(config-if)#int f0/0.201
msk-q42-arsenic-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.201, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.201, changed state to up

msk-q42-arsenic-gw-1(config-subif)#enc dot1q 201
msk-q42-arsenic-gw-1(config-subif)#ip addr 10.129.0.1 255.255.255.0
msk-q42-arsenic-gw-1(config-subif)#descr q42-main
msk-q42-arsenic-gw-1(config-subif)#int f1/0
msk-q42-arsenic-gw-1(config-if)#no sh

msk-q42-arsenic-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

msk-q42-arsenic-gw-1(config-if)#int f1/0.202
msk-q42-arsenic-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet1/0.202, changed state to down
```

Статическая маршрутизация в квартале 42

```
msk-q42-arsenio-gw-1(config-if)#int f1/0.202
msk-q42-arsenio-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet1/0.202, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0.202, changed state to up

msk-q42-arsenio-gw-1(config-subif)#enc dot1q 202
                                     ^
% Invalid input detected at '^' marker.

msk-q42-arsenio-gw-1(config-subif)#enc dot1q 202
msk-q42-arsenio-gw-1(config-subif)#ip addr 10.129.
                                     ^
% Invalid input detected at '^' marker.

msk-q42-arsenio-gw-1(config-subif)#ip addr 10.129.1.1 255.255.255.0
msk-q42-arsenio-gw-1(config-subif)#descr q42-management
msk-q42-arsenio-gw-1(config-subif)#exit
msk-q42-arsenio-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.128.255.1
msk-q42-arsenio-gw-1(config)#ip route 10.129.128.0 255.255.128.0 10.129.1.2
msk-q42-arsenio-gw-1(config)#^Z
msk-q42-arsenio-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-arsenio-gw-1#wr mem
Building configuration...
[OK]
msk-q42-arsenio-gw-1#
```

Рис. 5: Настройка статической маршрутизации на msk-q42-arsenio-gw-1

VLAN 201 на коммутаторе квартала 42

```
msk-q42-arsenio-sw-1>en
Password:
msk-q42-arsenio-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-arsenio-sw-1(config)#int f0/24
msk-q42-arsenio-sw-1(config-if)#sw mode trunk

msk-q42-arsenio-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

msk-q42-arsenio-sw-1(config-if)#int f0/1
msk-q42-arsenio-sw-1(config-if)#sw mode acc
msk-q42-arsenio-sw-1(config-if)#sw acc vlan 201
% Access VLAN does not exist. Creating vlan 201
msk-q42-arsenio-sw-1(config-if)#vlan 201
msk-q42-arsenio-sw-1(config-vlan)#name q42-main
msk-q42-arsenio-sw-1(config-vlan)#int vlan 201
msk-q42-arsenio-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan201, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan201, changed state to up

msk-q42-arsenio-sw-1(config-if)#no sh
msk-q42-arsenio-sw-1(config-if)#^Z
msk-q42-arsenio-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-arsenio-sw-1#wr mem
Building configuration...
[OK]
msk-q42-arsenio-sw-1#
```

Настройка сети общедоступности

```

Password:
msk-hostel-arsenio-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-hostel-arsenio-gw-1(config)#int g0/1
msk-hostel-arsenio-gw-1(config-if)#sw trunk enc dot1q
msk-hostel-arsenio-gw-1(config-if)#sw mode trunk

msk-hostel-arsenio-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

msk-hostel-arsenio-gw-1(config-if)#int f0/1
msk-hostel-arsenio-gw-1(config-if)#sw trunk enc dot1q
msk-hostel-arsenio-gw-1(config-if)#sw mode trunk
msk-hostel-arsenio-gw-1(config-if)#vlan 202
msk-hostel-arsenio-gw-1(config-vlan)#name q42-management
msk-hostel-arsenio-gw-1(config-vlan)#int vlan 202
msk-hostel-arsenio-gw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan202, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan202, changed state to up

msk-hostel-arsenio-gw-1(config-if)#no sh
msk-hostel-arsenio-gw-1(config-if)#ip addr 10.129.1.2 255.255.255.0
msk-hostel-arsenio-gw-1(config-if)#vlan 301
msk-hostel-arsenio-gw-1(config-vlan)#name hostel-main
msk-hostel-arsenio-gw-1(config-vlan)#int vlan 301
msk-hostel-arsenio-gw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan301, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up

msk-hostel-arsenio-gw-1(config-if)#no sh
msk-hostel-arsenio-gw-1(config-if)#ip addr 10.129.128.1 255.255.255.0
msk-hostel-arsenio-gw-1(config-if)#exit
msk-hostel-arsenio-gw-1(config)#ip routing
msk-hostel-arsenio-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.129.1.1
msk-hostel-arsenio-gw-1(config)#^Z
msk-hostel-arsenio-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-hostel-arsenio-gw-1#wr mem
Building configuration...
[OK]
msk-hostel-arsenio-gw-1#

```

```
msk-hostel-arsenio-sw-1>en
Password:
msk-hostel-arsenio-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-hostel-arsenio-sw-1(config)#
msk-hostel-arsenio-sw-1(config)#int g0/1
msk-hostel-arsenio-sw-1(config-if)#sw mode trunk
msk-hostel-arsenio-sw-1(config-if)#int f0/1
msk-hostel-arsenio-sw-1(config-if)#sw mode acc
msk-hostel-arsenio-sw-1(config-if)#sw acc vlan 301
% Access VLAN does not exist. Creating vlan 301
msk-hostel-arsenio-sw-1(config-if)#vlan 301
msk-hostel-arsenio-sw-1(config-vlan)#name hostel-main
msk-hostel-arsenio-sw-1(config-vlan)#int vlan 301
msk-hostel-arsenio-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan301, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up

msk-hostel-arsenio-sw-1(config-if)#no sh
msk-hostel-arsenio-sw-1(config-if)#^Z
msk-hostel-arsenio-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-hostel-arsenio-sw-1#wr mem
Building configuration...
[OK]
msk-hostel-arsenio-sw-1#
```

Рис. 8: Настройка VLAN 301 на msk-hostel-arsenio-sw-1

Настройка филиала в Сочи

Маршрутизатор sch-sochi-arsenio-gw-1

```

Password:
sch-sochi-arsenio-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
sch-sochi-arsenio-gw-1(config)#int f0/0
sch-sochi-arsenio-gw-1(config-if)#no sh

sch-sochi-arsenio-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

sch-sochi-arsenio-gw-1(config-if)#int f0/0.6
sch-sochi-arsenio-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.6, changed state to up

sch-sochi-arsenio-gw-1(config-subif)#enc dot1q 6
sch-sochi-arsenio-gw-1(config-subif)#ip addr 10.128.255.6 255.255.255.252
sch-sochi-arsenio-gw-1(config-subif)#descr donskaya
sch-sochi-arsenio-gw-1(config-subif)#int f0/0.401
sch-sochi-arsenio-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.401, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.401, changed state to up

sch-sochi-arsenio-gw-1(config-subif)#enc dot1q 401
sch-sochi-arsenio-gw-1(config-subif)#ip addr 10.130.0.1 255.255.255.0
sch-sochi-arsenio-gw-1(config-subif)#descr sochi-main
sch-sochi-arsenio-gw-1(config-subif)#int f0/0.402
sch-sochi-arsenio-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.402, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.402, changed state to up

sch-sochi-arsenio-gw-1(config-subif)#ip addr 10.130.1.1 255.255.255.0

% Configuring IP routing on a LAN subinterface is only allowed if that
subinterface is already configured as part of an IEEE 802.10, IEEE 802.1Q,
or ISL vLAN.

sch-sochi-arsenio-gw-1(config-subif)#enc dot1q 402
sch-sochi-arsenio-gw-1(config-subif)#ip addr 10.130.1.1 255.255.255.0
sch-sochi-arsenio-gw-1(config-subif)#descr sochi-management
sch-sochi-arsenio-gw-1(config-subif)#exit
sch-sochi-arsenio-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.128.255.5
sch-sochi-arsenio-gw-1(config)#^Z
sch-sochi-arsenio-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
```

Настройка оконечных устройств

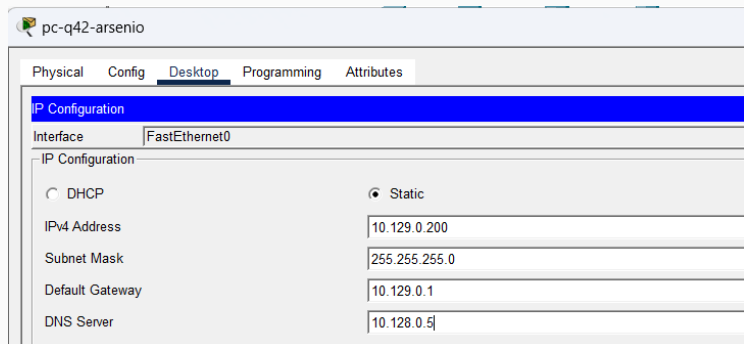


Рис. 10: Настройка IP-адреса на pc-q42-arsenio

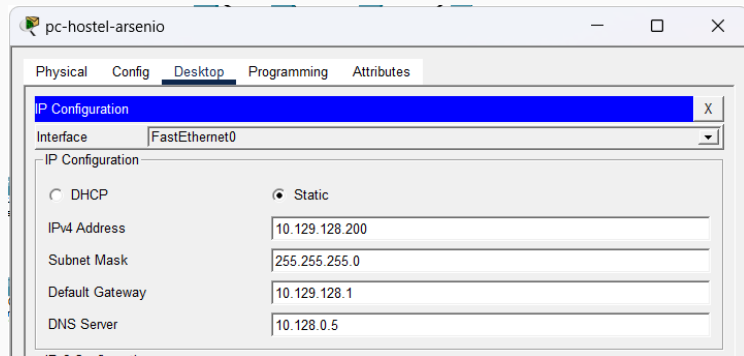


Рис. 11: Настройка IP-адреса на pc-hostel-arsenio

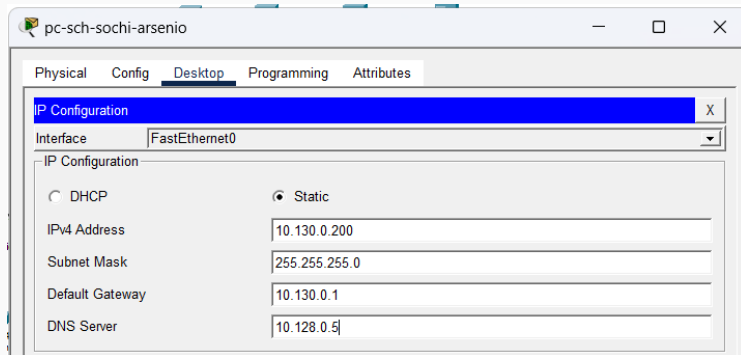
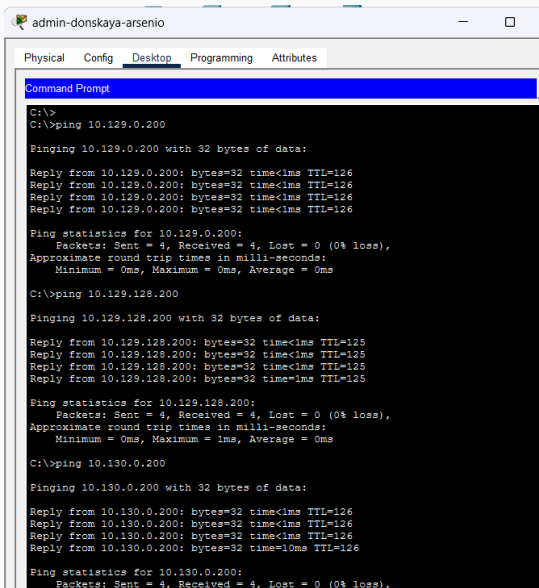


Рис. 12: Настройка IP-адреса на pc-sch-sochi-arsenio

Проверка связности

Ping c admin-donskaya-arsenio



```
admin-donskaya-arsenio
Physical Config Desktop Programming Attributes
Command Prompt
C:\>
C:\>ping 10.129.0.200

Pinging 10.129.0.200 with 32 bytes of data:

Reply from 10.129.0.200: bytes=32 time<1ms TTL=126
Reply from 10.129.0.200: bytes=32 time<1ms TTL=126
Reply from 10.129.0.200: bytes=32 time<1ms TTL=126
Reply from 10.129.0.200: bytes=32 time<1ms TTL=126

Ping statistics for 10.129.0.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.129.128.200

Pinging 10.129.128.200 with 32 bytes of data:

Reply from 10.129.128.200: bytes=32 time<1ms TTL=125
Reply from 10.129.128.200: bytes=32 time<1ms TTL=125
Reply from 10.129.128.200: bytes=32 time<1ms TTL=125
Reply from 10.129.128.200: bytes=32 time=1ms TTL=125

Ping statistics for 10.129.128.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

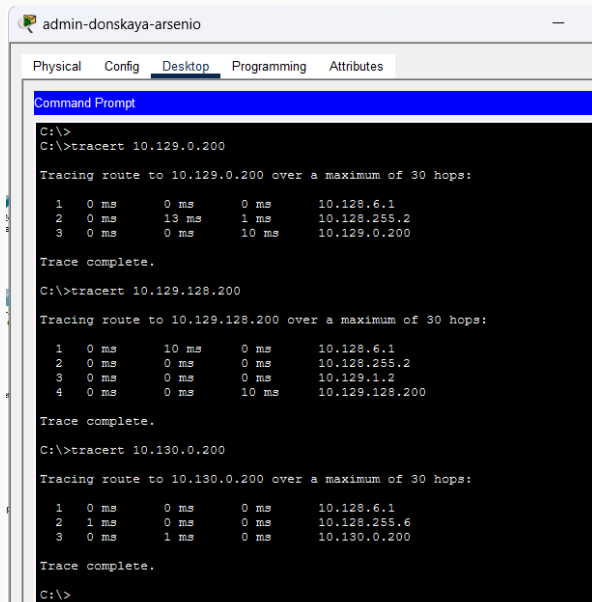
C:\>ping 10.130.0.200

Pinging 10.130.0.200 with 32 bytes of data:

Reply from 10.130.0.200: bytes=32 time<1ms TTL=126
Reply from 10.130.0.200: bytes=32 time<1ms TTL=126
Reply from 10.130.0.200: bytes=32 time<1ms TTL=126
Reply from 10.130.0.200: bytes=32 time=10ms TTL=126

Ping statistics for 10.130.0.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

Tracert c admin-donskaya-arsenio



The screenshot shows a Windows desktop environment. At the top, there is a taskbar with a single application window titled "admin-donskaya-arsenio". The window has four tabs: "Physical", "Config", "Desktop" (which is selected), and "Programming". Below the tabs is a "Command Prompt" window. The Command Prompt displays the following text:

```
C:\>
C:\>tracert 10.129.0.200

Tracing route to 10.129.0.200 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.128.6.1
  2  0 ms    13 ms   1 ms    10.128.255.2
  3  0 ms     0 ms   10 ms   10.129.0.200

Trace complete.

C:\>tracert 10.129.128.200

Tracing route to 10.129.128.200 over a maximum of 30 hops:

  1  0 ms     10 ms   0 ms    10.128.6.1
  2  0 ms     0 ms    0 ms    10.128.255.2
  3  0 ms     0 ms    0 ms    10.129.1.2
  4  0 ms     0 ms   10 ms   10.129.128.200

Trace complete.

C:\>tracert 10.130.0.200

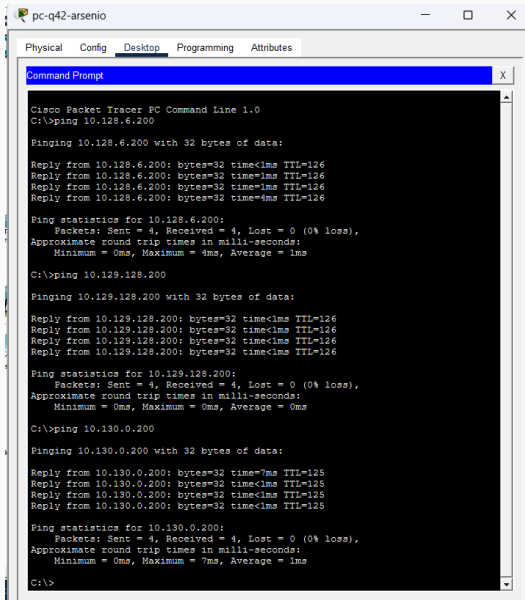
Tracing route to 10.130.0.200 over a maximum of 30 hops:

  1  0 ms     0 ms    0 ms    10.128.6.1
  2  1 ms     0 ms    0 ms    10.128.255.6
  3  0 ms     1 ms    0 ms    10.130.0.200

Trace complete.

C:\>
```


Ping c pc-q42-arsenio



```
pc-q42-arsenio
Physical Config Desktop Programming Attributes
Command Prompt X

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.6.200

Pinging 10.128.6.200 with 32 bytes of data:

Reply from 10.128.6.200: bytes=32 time<1ms TTL=126
Reply from 10.128.6.200: bytes=32 time<1ms TTL=126
Reply from 10.128.6.200: bytes=32 time<1ms TTL=126
Reply from 10.128.6.200: bytes=32 time=4ms TTL=126

Ping statistics for 10.128.6.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>ping 10.129.128.200

Pinging 10.129.128.200 with 32 bytes of data:

Reply from 10.129.128.200: bytes=32 time<1ms TTL=126
Reply from 10.129.128.200: bytes=32 time<1ms TTL=126
Reply from 10.129.128.200: bytes=32 time<1ms TTL=126
Reply from 10.129.128.200: bytes=32 time<1ms TTL=126

Ping statistics for 10.129.128.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.130.0.200

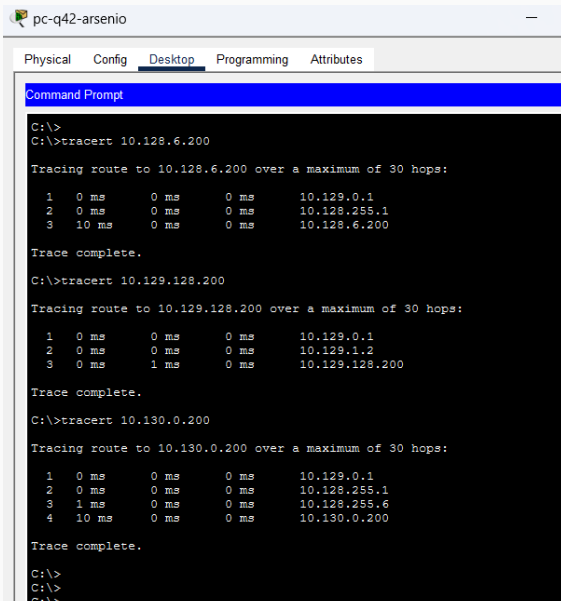
Pinging 10.130.0.200 with 32 bytes of data:

Reply from 10.130.0.200: bytes=32 time=7ms TTL=125
Reply from 10.130.0.200: bytes=32 time<1ms TTL=125
Reply from 10.130.0.200: bytes=32 time<1ms TTL=125
Reply from 10.130.0.200: bytes=32 time<1ms TTL=125

Ping statistics for 10.130.0.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 1ms

C:\>
```

Tracert c pc-q42-arsenio



```
C:\>
C:\>tracert 10.128.6.200

Tracing route to 10.128.6.200 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.129.0.1
  2  0 ms    0 ms    0 ms    10.128.255.1
  3  10 ms   0 ms    0 ms    10.128.6.200

Trace complete.

C:\>tracert 10.129.128.200

Tracing route to 10.129.128.200 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.129.0.1
  2  0 ms    0 ms    0 ms    10.129.1.2
  3  0 ms    1 ms    0 ms    10.129.128.200

Trace complete.

C:\>tracert 10.130.0.200

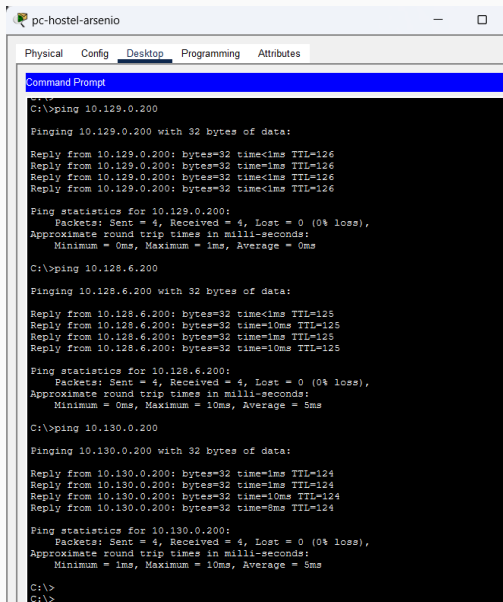
Tracing route to 10.130.0.200 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.129.0.1
  2  0 ms    0 ms    0 ms    10.128.255.1
  3  1 ms    0 ms    0 ms    10.128.255.6
  4  10 ms   0 ms    0 ms    10.130.0.200

Trace complete.

C:\>
C:\>
C:\>
```

Ping c pc-hostel-arsenio



```
pc-hostel-arsenio
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 10.129.0.200

Pinging 10.129.0.200 with 32 bytes of data:

Reply from 10.129.0.200: bytes=32 time<1ms TTL=126
Reply from 10.129.0.200: bytes=32 time<1ms TTL=126
Reply from 10.129.0.200: bytes=32 time<1ms TTL=126
Reply from 10.129.0.200: bytes=32 time<1ms TTL=126

Ping statistics for 10.129.0.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.128.6.200

Pinging 10.128.6.200 with 32 bytes of data:

Reply from 10.128.6.200: bytes=32 time<1ms TTL=125
Reply from 10.128.6.200: bytes=32 time=10ms TTL=125
Reply from 10.128.6.200: bytes=32 time=1ms TTL=125
Reply from 10.128.6.200: bytes=32 time=10ms TTL=125

Ping statistics for 10.128.6.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 5ms

C:\>ping 10.130.0.200

Pinging 10.130.0.200 with 32 bytes of data:

Reply from 10.130.0.200: bytes=32 time=1ms TTL=124
Reply from 10.130.0.200: bytes=32 time=1ms TTL=124
Reply from 10.130.0.200: bytes=32 time=10ms TTL=124
Reply from 10.130.0.200: bytes=32 time=8ms TTL=124

Ping statistics for 10.130.0.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 10ms, Average = 5ms

C:\>
C:\>
```

```
C:\>
C:\>tracert 10.129.0.200

Tracing route to 10.129.0.200 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      10.129.128.1
  2  0 ms      0 ms      0 ms      10.129.1.1
  3  1 ms      0 ms      1 ms      10.129.0.200

Trace complete.

C:\>tracert 10.128.6.200

Tracing route to 10.128.6.200 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      10.129.128.1
  2  0 ms      0 ms      0 ms      10.129.1.1
  3  2 ms      0 ms      1 ms      10.128.255.1
  4  0 ms      0 ms      0 ms      10.128.6.200

Trace complete.

C:\>tracert 10.130.0.200

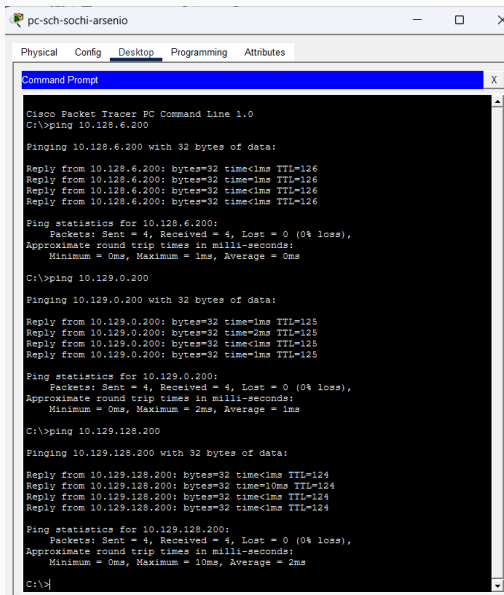
Tracing route to 10.130.0.200 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      10.129.128.1
  2  0 ms      0 ms      1 ms      10.129.1.1
  3  0 ms      0 ms      0 ms      10.128.255.1
  4  0 ms      0 ms      0 ms      10.128.255.6
  5  0 ms      0 ms      0 ms      10.130.0.200

Trace complete.

C:\>
```

Ping c pc-sch-sochi-arsenio



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'pc-sch-sochi-arsenio'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' currently selected. The Command Prompt shows the following commands and output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.6.200

Pinging 10.128.6.200 with 32 bytes of data:

Reply from 10.128.6.200: bytes=32 time<1ms TTL=126
Reply from 10.128.6.200: bytes=32 time<1ms TTL=126
Reply from 10.128.6.200: bytes=32 time<1ms TTL=126
Reply from 10.128.6.200: bytes=32 time<1ms TTL=126

Ping statistics for 10.128.6.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.129.0.200

Pinging 10.129.0.200 with 32 bytes of data:

Reply from 10.129.0.200: bytes=32 time=1ms TTL=125
Reply from 10.129.0.200: bytes=32 time=2ms TTL=125
Reply from 10.129.0.200: bytes=32 time<1ms TTL=125
Reply from 10.129.0.200: bytes=32 time=1ms TTL=125

Ping statistics for 10.129.0.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 1ms

C:\>ping 10.129.128.200

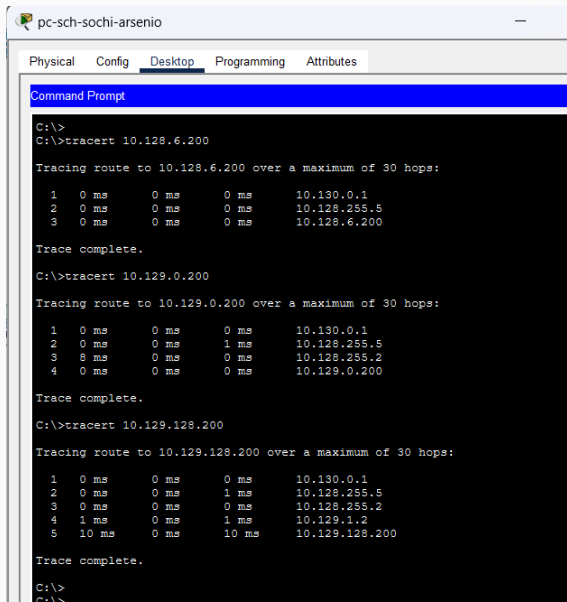
Pinging 10.129.128.200 with 32 bytes of data:

Reply from 10.129.128.200: bytes=32 time<1ms TTL=124
Reply from 10.129.128.200: bytes=32 time=10ms TTL=124
Reply from 10.129.128.200: bytes=32 time<1ms TTL=124
Reply from 10.129.128.200: bytes=32 time<1ms TTL=124

Ping statistics for 10.129.128.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>|
```

Tracert c pc-sch-sochi-arsenio



The screenshot shows a Windows desktop with a single window titled "pc-sch-sochi-arsenio". The window has four tabs: "Physical", "Config", "Desktop", and "Attributes". The "Desktop" tab is active. Inside the window is a "Command Prompt" window with a black background and white text. The text shows three consecutive tracert commands being executed from the C:\ prompt. Each command shows the route to a specific IP address, the number of hops, and the time taken at each hop.

```
C:\>
C:\>tracert 10.128.6.200

Tracing route to 10.128.6.200 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.130.0.1
  2  0 ms    0 ms    0 ms    10.128.255.5
  3  0 ms    0 ms    0 ms    10.128.6.200

Trace complete.

C:\>tracert 10.129.0.200

Tracing route to 10.129.0.200 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.130.0.1
  2  0 ms    0 ms    1 ms    10.128.255.5
  3  8 ms    0 ms    0 ms    10.128.255.2
  4  0 ms    0 ms    0 ms    10.129.0.200

Trace complete.

C:\>tracert 10.129.128.200

Tracing route to 10.129.128.200 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.130.0.1
  2  0 ms    0 ms    1 ms    10.128.255.5
  3  0 ms    0 ms    0 ms    10.128.255.2
  4  1 ms    0 ms    1 ms    10.129.1.2
  5  10 ms   0 ms    10 ms   10.129.128.200

Trace complete.

C:\>
C:\>
```

Таблицы маршрутизации

Таблица маршрутизации msk-donskaya-arsenio-gw-1

```
msk-donskaya-arsenio-gw-1#sh ip rou
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 198.51.100.1 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 18 subnets, 4 masks
C    10.128.0.0/24 is directly connected, FastEthernet0/0.3
L    10.128.0.1/32 is directly connected, FastEthernet0/0.3
C    10.128.1.0/24 is directly connected, FastEthernet0/0.2
L    10.128.1.1/32 is directly connected, FastEthernet0/0.2
C    10.128.3.0/24 is directly connected, FastEthernet0/0.101
L    10.128.3.1/32 is directly connected, FastEthernet0/0.101
C    10.128.4.0/24 is directly connected, FastEthernet0/0.102
L    10.128.4.1/32 is directly connected, FastEthernet0/0.102
C    10.128.5.0/24 is directly connected, FastEthernet0/0.103
L    10.128.5.1/32 is directly connected, FastEthernet0/0.103
C    10.128.6.0/24 is directly connected, FastEthernet0/0.104
L    10.128.6.1/32 is directly connected, FastEthernet0/0.104
C    10.128.255.0/30 is directly connected, FastEthernet0/1.5
L    10.128.255.1/32 is directly connected, FastEthernet0/1.5
C    10.128.255.4/30 is directly connected, FastEthernet0/1.6
L    10.128.255.5/32 is directly connected, FastEthernet0/1.6
S    10.129.0.0/16 [1/0] via 10.128.255.2
S    10.130.0.0/16 [1/0] via 10.128.255.6
198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C    198.51.100.0/28 is directly connected, FastEthernet0/1.4
L    198.51.100.2/32 is directly connected, FastEthernet0/1.4
S*   0.0.0.0/0 [1/0] via 198.51.100.1

msk-donskaya-arsenio-gw-1#
```


Таблица маршрутизации msk-q42-arsenio-gw-1

```
msk-q42-arsenio-gw-1#
msk-q42-arsenio-gw-1#sh ip rou
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.128.255.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 7 subnets, 4 masks
C       10.128.255.0/30 is directly connected, FastEthernet0/1.5
L       10.128.255.2/32 is directly connected, FastEthernet0/1.5
C       10.129.0.0/24 is directly connected, FastEthernet0/0.201
L       10.129.0.1/32 is directly connected, FastEthernet0/0.201
C       10.129.1.0/24 is directly connected, FastEthernet1/0.202
L       10.129.1.1/32 is directly connected, FastEthernet1/0.202
S       10.129.128.0/17 [1/0] via 10.129.1.2
S*    0.0.0.0/0 [1/0] via 10.128.255.1

msk-q42-arsenio-gw-1#
```

Рис. 22: Таблица маршрутизации на msk-q42-arsenio-gw-1

Таблица маршрутизации msk-hostel-arsenio-gw-1

```
msk-hostel-arsenio-gw-1#sh ip rou
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.129.1.1 to network 0.0.0.0

    10.0.0.0/24 is subnetted, 2 subnets
C      10.129.1.0 is directly connected, Vlan202
C      10.129.128.0 is directly connected, Vlan301
S*    0.0.0.0/0 [1/0] via 10.129.1.1
msk-hostel-arsenio-gw-1#
```

Рис. 23: Таблица маршрутизации на msk-hostel-arsenio-gw-1

Таблица маршрутизации sch-sochi-arsenio-gw-1

```
sch-sochi-arsenio-gw-1#sh ip rou
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.128.255.5 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
C       10.128.255.4/30 is directly connected, FastEthernet0/0.6
L       10.128.255.6/32 is directly connected, FastEthernet0/0.6
C       10.130.0.0/24 is directly connected, FastEthernet0/0.401
L       10.130.0.1/32 is directly connected, FastEthernet0/0.401
C       10.130.1.0/24 is directly connected, FastEthernet0/0.402
L       10.130.1.1/32 is directly connected, FastEthernet0/0.402
S*     0.0.0.0/0 [1/0] via 10.128.255.5

sch-sochi-arsenio-gw-1#
sch-sochi-arsenio-gw-1#
```

Рис. 24: Таблица маршрутизации на sch-sochi-arsenio-gw-1

Выводы

В ходе лабораторной работы была настроена связь между несколькими территориями организации в Cisco Packet Tracer.

Были созданы VLAN, настроены сабинтерфейсы маршрутизаторов, добавлены статические маршруты между территориями и подготовлена настройка NAT.

Проверка с помощью команд ping, tracert и show ip route подтвердила корректную работу маршрутизации между кварталом 42, сетью общежития, территорией Донской и филиалом в г. Сочи.